# Quantum Systems

(Lecture 4: Quantum gates and the circuit model)

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# The circuit model

Classical reversible circuits (which can simulate any non-reversible one with modest overhead) generalise to quantum circuits where

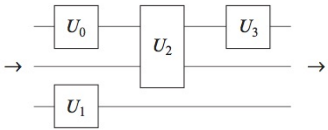
- logical qubits are carried along wires,
- quantum gates, corresponding to unitary transformations, act on them,
- and measurements of a quantum state $|\phi\rangle = \sum_i \alpha_i |i\rangle$result in a state $|i\rangle$, with probability given by the norm squared of its amplitude, $\| \alpha_i \|^2$, together with a classical label $i$ indicating which outcome was obtained.

# The circuit model

## Quantum gates

A gate is a transformation that acts on only a small number of qubits
Differently from the classical case, they do not necessarily correspond to physical objects

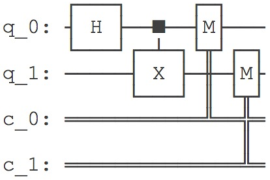
## Notation



# The circuit model

## Circuits in Qiskit

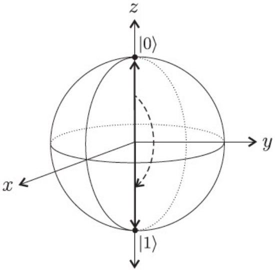
```
from qiskit import QuantumCircuit

qc = QuantumCircuit(2, 2)
qc.h(0)
qc.cx(0, 1)
qc.measure([0, 1], [0, 1])
qc.draw()
```



# 1-Gates

The action of a 1-gate $U$ on a quantum state $|\phi\rangle$ can be thought of as a rotation of the Bloch vector for $|\phi\rangle$ to the Bloch vector for $U|\phi\rangle$, eg.

## Example: $X$




is a rotation about the $x$ axis.

# 1-Gates

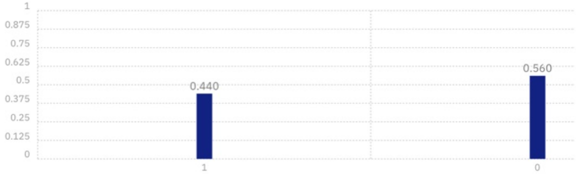
The $X = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$ gate


$$X|0\rangle = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 0 \\ 1 \end{bmatrix} = |1\rangle$$

# 1-Gates

## The Hadamard gate creates superpositions

$$H = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}$$




$$H|0\rangle = |+\rangle = \overbrace{\frac{1}{\sqrt{2}}(|0\rangle + |1\rangle)}^{\text{superposition}}$$

$$H|1\rangle = |-\rangle = \frac{1}{\sqrt{2}}(|0\rangle - |1\rangle)$$

# 1-Gates

## The phase shift gate

$$R_\phi = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 0 \\ 0 & e^{i\phi} \end{bmatrix}$$

$$\begin{aligned} R_\phi \left|0\right\rangle &= \left|0\right\rangle \\ R_\phi \left|1\right\rangle &= e^{i\phi}\left|1\right\rangle \end{aligned}$$

## The $T$ (or $\frac{\pi}{8}$) gate

$$T = R_{\frac{\pi}{4}} = \begin{bmatrix} 1 & 0 \\ 0 & e^{i\frac{\pi}{4}} \end{bmatrix}$$

which, up to a global phase factor $e^{i\frac{\pi}{8}}$, is equivalent to

$$\begin{bmatrix} e^{-i\frac{\pi}{8}} & 0 \\ 0 & e^{i\frac{\pi}{8}} \end{bmatrix}$$

# 1-Gates

## Pauli gates

$$
\begin{aligned}
I &= |0\rangle\langle 0| + |1\rangle\langle 1| = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \\
X &= |1\rangle\langle 0| + |0\rangle\langle 1| = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \\
Z &= |0\rangle\langle 0| - |1\rangle\langle 1| = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} = R_\pi \\
Y &= i(-|1\rangle\langle 0| + |0\rangle\langle 1|) = \begin{bmatrix} 0 & -i \\ i & 0 \end{bmatrix}
\end{aligned}
$$

# 1-Gates

## Rotation gates

Correspond to rotations about the three axes of the Bloch sphere, and are computed as Pauli gates squared.

$$R_e(\theta) \mathrel{\widehat{=}} e^{\frac{-i\theta E}{2}} = \cos\left(\frac{\theta}{2}\right) I - i \sin \frac{\theta}{2} E$$

where $e \mathrel{\widehat{=}} x, y, z$ and $E \mathrel{\widehat{=}} X, Y, Z$.

because, for any real number $r$ and matrix $R$ st $R^2 = I$, which is the case for $X$, $Y$, and $Z$,

$$e^{irR} = cos(r) I + i \sin(r) R$$

# 1-Gates

## Rotation gates as matrices in the computational basis

$$R_x(\theta) = \begin{bmatrix} \cos\left(\frac{\theta}{2}\right) & -i\sin\left(\frac{\theta}{2}\right) \\ -i\sin\left(\frac{\theta}{2}\right) & \cos\left(\frac{\theta}{2}\right) \end{bmatrix}$$

$$R_y(\theta) = \begin{bmatrix} \cos\left(\frac{\theta}{2}\right) & -\sin\left(\frac{\theta}{2}\right) \\ \sin\left(\frac{\theta}{2}\right) & \cos\left(\frac{\theta}{2}\right) \end{bmatrix}$$

$$R_z(\theta) = \begin{bmatrix} e^{-i\frac{\theta}{2}} & 0 \\ 0 & e^{i\frac{\theta}{2}} \end{bmatrix}$$

# 1-Gates

Compute $R_z(\theta)|\psi\rangle$ for $|\psi\rangle = \cos\left(\frac{\sigma}{2}\right)|0\rangle + e^{i\gamma}\sin\left(\frac{\sigma}{2}\right)|1\rangle$

$$
\begin{aligned}
\begin{bmatrix} e^{-i\frac{\theta}{2}} & 0 \\ 0 & e^{i\frac{\theta}{2}} \end{bmatrix}
\begin{bmatrix} \cos\left(\frac{\sigma}{2}\right) \\ e^{i\gamma}\sin\left(\frac{\sigma}{2}\right) \end{bmatrix}
&= \begin{bmatrix} e^{-i\frac{\theta}{2}}\cos\left(\frac{\sigma}{2}\right) \\ e^{i\frac{\theta}{2}}e^{i\gamma}\sin\left(\frac{\sigma}{2}\right) \end{bmatrix} \\
&= e^{-i\frac{\theta}{2}} \begin{bmatrix} \cos\left(\frac{\sigma}{2}\right) \\ e^{i\theta}e^{i\gamma}\sin\left(\frac{\sigma}{2}\right) \end{bmatrix} \\
&= e^{-i\frac{\theta}{2}} \left(\cos\left(\frac{\sigma}{2}\right)|0\rangle + e^{i(\gamma+\theta)}\sin\left(\frac{\sigma}{2}\right)|1\rangle\right)
\end{aligned}
$$

As global phase is insignificant, the angle mapping $\gamma \mapsto \gamma + \theta$ is a rotation of $\theta$ around the $z$-axis of the Bloch sphere.

# 1-Gates

### Theorem

Let $U$ be a 1-gate, and $v, w$ any two non-parallel axes of the Bloch sphere. Then there exist real numbers $\alpha, \beta\, \gamma, \delta$ st

$$U = e^{i\alpha} R_v(\beta) R_w(\gamma) R_v(\delta)$$

which means that any 1-gate can be expressed as a sequence of two rotations about an axis and one rotation about another non parallel axis, multiplied by a suitable phase factor.

proof hint: Recall $U$ is unitary and unfold the definition of rotation gate.

# 2-gates: *CNOT*

Acts on the standard basis for a 2-qubit system, flipping the second bit if the first bit is 1 and leaving it unchanged otherwise.

$$
\begin{aligned}
CNOT &= |0\rangle\langle 0| \otimes I \;+\; |1\rangle\langle 1| \otimes X \\
&= |0\rangle\langle 0| \otimes (|0\rangle\langle 0| + |1\rangle\langle 1|) \;+\; |1\rangle\langle 1| \otimes (|1\rangle\langle 0| + |0\rangle\langle 1|) \\
&= |00\rangle\langle 00| + |01\rangle\langle 01| + |11\rangle\langle 10| + |10\rangle\langle 11| \\
&= \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{bmatrix}
\end{aligned}
$$

*CNOT* is unitary and is its own inverse, and cannot be decomposed into a tensor product of two 1-qubit transformations

# 2-gates: *CNOT*

The importance of *CNOT* is its ability to change the entanglement between two qubits, e.g.

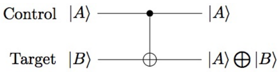
$$\begin{aligned} CNOT\left(\frac{1}{\sqrt{2}}(|0\rangle + |1\rangle) \otimes |0\rangle\right) &= CNOT\left(\frac{1}{\sqrt{2}}(|00\rangle + |10\rangle)\right) \\ &= \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle) \end{aligned}$$

Being its own inverse, also takes an entangled state to an unentangled one.

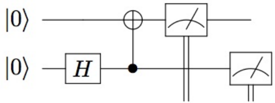
Note that entanglement is not a local property in the sense that transformations that act separately on two or more subsystems cannot affect the entanglement between those subsystems:

$$(U \otimes V)\,|v\rangle \quad \text{is entangled} \quad \text{iff} \quad |v\rangle \text{ is}$$

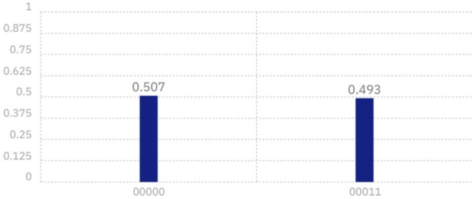
# 2-gates: *CNOT*



Control $|A\rangle$
Target $|B\rangle$
$|A\rangle$
$|A\rangle \oplus |B\rangle$

$|0\rangle$
$|0\rangle$
$H$

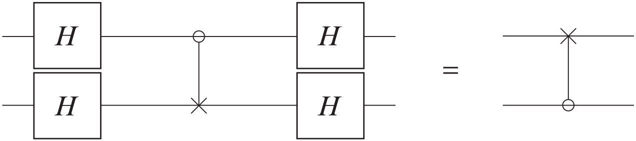
1
0.875
0.75
0.625
0.5
0.375
0.25
0.125
0
0.507
0.493
00000
00011

# 2-gates: *CNOT*

The notions of control/target bit in *CNOT* are arbitrary: they depend on what basis is considered. The standard behaviour is obtained in the computational basis. However, roles are interchanged in the Hadamard basis in which the effect of *CNOT* is

$$|++\rangle \mapsto |++\rangle \quad |+-\rangle \mapsto |--\rangle \quad |-+\rangle \mapsto |-+\rangle \quad |--\rangle \mapsto |+-\rangle$$

## Exercise



# The proof

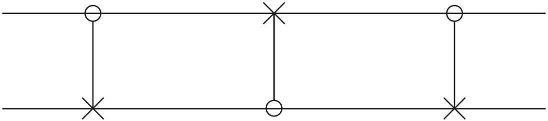
$$
\begin{aligned}
LHS &= \frac{1}{2}\begin{bmatrix} H & H \\ H & -H \end{bmatrix} \overbrace{\begin{bmatrix} I & 0 \\ 0 & X \end{bmatrix}}^{CNOT} \begin{bmatrix} H & H \\ H & -H \end{bmatrix} \\
&= \frac{1}{2}\begin{bmatrix} H & HX \\ H & -HX \end{bmatrix} \begin{bmatrix} H & H \\ H & -H \end{bmatrix} \\
&= \frac{1}{2}\begin{bmatrix} I + HXH & I - HXH \\ I - HXH & I + HXH \end{bmatrix} \quad = \quad \frac{1}{2}\begin{bmatrix} I + Z & I - Z \\ I - Z & I + Z \end{bmatrix} \\
&= \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \end{bmatrix} \\
&= I \otimes |0\rangle\langle 0| + X \otimes |1\rangle\langle 1| \quad = \quad RHS
\end{aligned}
$$

noting that

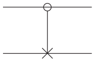
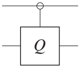
$$
H \otimes H = (I \otimes H)(H \otimes I) = \frac{1}{\sqrt{2}}\begin{bmatrix} H & 0 \\ 0 & H \end{bmatrix} \begin{bmatrix} I & I \\ I & -I \end{bmatrix} = \frac{1}{\sqrt{2}}\begin{bmatrix} H & H \\ H & -H \end{bmatrix}
$$

# Exercise

Discuss



# Controlled $Q$-gates

From  to 


$$
\begin{aligned}
C_Q|0\rangle|\varphi\rangle &= |0\rangle|\varphi\rangle \\
C_Q|1\rangle|\varphi\rangle &= |1\rangle Q|\varphi\rangle
\end{aligned}
$$

$$C_Q = |0\rangle\langle 0| \otimes I + |1\rangle\langle 1| \otimes Q$$

corresponding to the following matrix in the standard basis:

$$C_Q = \begin{bmatrix} 1 & 0 \\ 0 & Q \end{bmatrix}$$

# Controlled phase shift gate

$$C_{e^{i\theta}} \;=\; |00\rangle\langle 00| + |01\rangle\langle 01| + e^{i\theta}|10\rangle\langle 10| + e^{i\theta}|11\rangle\langle 11|$$

$$C_{e^{i\theta}} \;=\; \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & e^{i\theta} & 0 \\ 0 & 0 & 0 & e^{i\theta} \end{bmatrix}$$

## Transforming a global into a local phase

$$\frac{1}{\sqrt{2}}(|00\rangle + |11\rangle) \quad \longrightarrow \quad \frac{1}{\sqrt{2}}(|00\rangle + e^{i\theta}|11\rangle)$$

Actually, a unitary transformation is completely determined by its action on a basis, but not by specifying what states the states corresponding to basis states are sent to.

Example: $e^{i\theta}$ takes the four quantum states to themselves (because e.g. $|10\rangle$ and $e^{i\theta}|10\rangle$ represent the same state), but a global phase can be transformed into a local one, as above

# *CCNOT* or Toffoli gate

A 3-bit gate corresponding to controlled *CNOT*. If the first two bits are in the state $|1\rangle$ applies $X$ the third bit, else it does nothing:

$$|q_1 q_2 q_3\rangle \mapsto |q_1 q_2, q_3 \oplus (q_1 \wedge q_2)\rangle$$

In matrix form,

$$\begin{bmatrix} 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \end{bmatrix}$$

# Universal set of gates?

## Is there a universal set of quantum gates?

In general no: there are uncountably many quantum transformations, and a finite set of generators can only generate countably many elements. However, it is possible for finite sets of gates to generate arbitrarily close approximations to all unitary transformations.

## Definitions

- The error in approximating $U$ by $V$ is

$$Er(U, V) \;=\; \max_{|\phi\rangle} \; \|\, (U - V)|\phi\rangle \,\|$$

- An operator $U$ can be approximated to arbitrary accuracy if for any positive $\epsilon$ there exists another unitary transformation $V$ st $Er(U, V) \leq \epsilon$.
- A set of gates is universal if for any integer $n \geq 1$, any $n$-qubit unitary operator can be approximated to arbitrary accuracy by a quantum circuit using only gates from that set.

# Universal set of gates?

## Some examples

- The set $\{H, T\}$ is universal for 1-gates.
- The set $\{H, T, CNOT\}$ is a universal set of gates.

## How efficient is an approximation?

To approximate an unitary transformation encoding some specific computation, one would expect to use a number of gates from the universal set which is polynomial in the number of qubits and the inverse of the quality factor $\epsilon$.

Main result: theorem of Solovay-Kitaev